

1 Configuration of Centralized MDE

Purpose

You can use the configuration of the centralized MDE to collect machine data using a Process Communication Controller (PCC) installed stand-alone. The following collection types are supported:

- Machine monitoring based on cycles or operating signals
- Collection of cycles or machine statuses.

Requirements

The centralized MDE is part of a product and requires a license.

To use the centralized MDE, the following software requirements must be fulfilled:

- Maintenance Manager 2, as of version 2.0.2881, is installed on the HYDRA server.
- Messaging protocol MQTT (hymqtt.dll/.so), as of version 8.1.1.0 is installed (the installation steps are described in section *Installation MQTT Broker*).

Additionally, the following programs are required:

- HYDRA 8 service pack 14 is installed.
We recommend to install the up-to-date software version.
- MES Weaver (hymw.exe/.out) as of version 8.1.1.619
- PCC (pcc.exe) as of version 7.2.3.3
- MDE blade 2 (mdeb2.dll) as of version 8.1.2.1
- AIP 8.2 (ctaip.exe) as of version 8.2.1.32.
- On the AIP/PCC, the "Visual C++ Redistributable Package for Visual Studio 2010 x86" must be installed.
- On the AIP/PCC, the "Visual C++ Redistributable Package for Visual Studio 2013 x86" must be installed.

You must have run the database patch *dbp_persistent_productionlock.hsc* on the HYDRA server.

Windows: `hydscr.exe db_sql\dbp_persistent_productionlock.hsc`

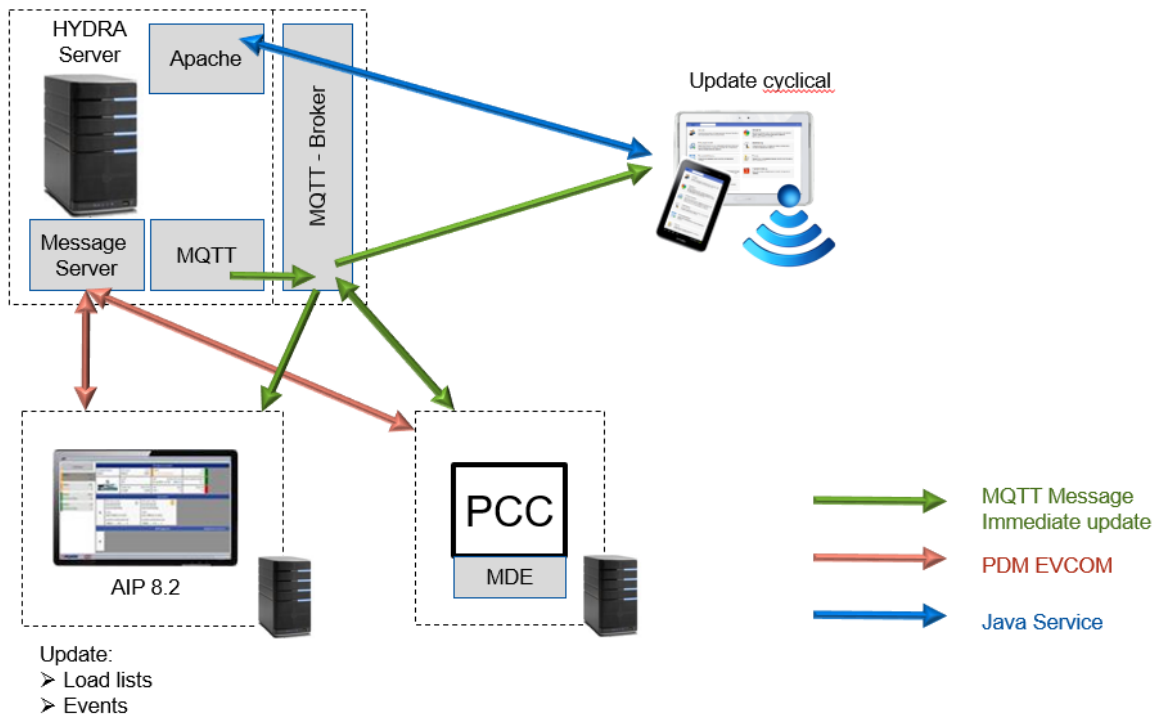
Linux: `hydscr.out db_sql\dbp_persistent_productionlock.hsc`



For the stand-alone installation of the PCC, a computer must be available that does **not** run AIP 8.2. For this computer, the same hardware and software recommendations apply as for the AIP 8.2.

Structure of a centralized MDE

The diagram below illustrates the structure. The arrows are the communication protocols used.



Note for the centralized machine data collection (stand-alone PCC)



If you configure the centralized MDE, different communication technologies are in use. As a result, the processing logic of the machine data collection is completely different to the one in an AIP (where the PCC runs in combined operation).

If you use the centralized MDE, the so-called combined operation (the PCC runs in the same terminal number as the AIP) is *not* supported.

In addition, note the following:

- If machines are monitored automatically, monitoring may only be performed by the PCC.
- If quantities are recorded automatically, automatic quantities may only be recorded by the PCC.
- When assigning a workplace to (further) terminals (AIP 8.2), the number of terminals should be kept to a minimum. The communication between terminals increases significantly and may have a negative effect on the availability of terminals.
- Workplaces assigned to several terminals must not have a workplace/machine status configured with the control indicator "no order".
- The list of batches logged on to the AIP 8.2 ("3rd list") is not part of the synchronization function.
- The function "Entry of disturbance reason required" is not supported.

- The posting volume and the posting frequency in the terminals, especially during a shift change, is important: in a worst-case scenario, there might be problems with synchronization and thus incorrect postings.
- Customizations are *not* compatible by default. We recommend to test this in the run-up.
- If workplaces are assigned to several terminals, you may only assign one shop floor terminal (AIP 8.2) as MDE terminal.
- The function "Set production lock" is only available on terminals with the configuration "operated as MDE terminal" (up to CTAIP version 8.2.1.34).

Installation MQTT Broker

You need to install the MQTT Broker in one system only once.



[The installation instruction of the MQTT Browser is included in the MES Weaver 4.0_{pe} installation instruction](#) (only available in English).

Configuration of the "stand-alone PCC"

Configure as follows:

MOC: INI data configuration

Make the following [INI data configurations](#) on the MOC for each workplace that is assigned to a stand-alone PCC. With this configuration, the HYDRA server notifies the PCC each time the server receives a posting for the workplace.

Name	MDE
Section	MQTT_NOTIFICATION
Key	<workplace number according to configuration>
Value	TRUE
Active	☒
Comment (optional)	MQTT notification to PCC for defined workplace

MOC: INI data configuration

If the PCC performs machine monitoring plus counter recording, also make the following [INI data configurations](#):

Name	MDE
Section	GET_COUNTER_FROM_MDE_TERMINAL
Key	ACTIVE
Value	TRUE
Active	☒
Comment (optional)	

Name	MDE
Section	GET_COUNTER_FROM_MDE_TERMINAL
Key	SHIFT_THRESHOLD
Value	300
Active	☒
Comment (optional)	

The processing logic of these options is described in the document [MBL_Distributed_MDE_Counter.pdf](#).

MOC: Terminal configuration

Configure the PCC in the [Terminal configuration](#) as terminal of terminal type "PCC" (141). Enable the options "Operated as BDE terminal" and "Operated as MDE terminal".

MOC: Assignment of workplaces/machines to terminals

Workplaces for which the PCC records signals: Assign the workplaces to the PCC in the [Workplace terminal assignment](#).



You can assign a maximum of 16 workplaces/machines to a PCC. Configure further PCCs as terminals, if you want to record data for more than 16 workplaces via the centralized MDE.

Installation of the PCC as stand-alone application

For information on the configuration of the PCC as service, refer to section 3 of the documentation [SCS-PCB_30.pdf](#).



Do **not** install the PCC in a hardware running an AIP 8.2. Use a separate hardware.

Using the compatibility mode

With some operating systems, it is possible that the PCC finishes unexpectedly when using MQTT (centralized MDE). To avoid this behavior, run the stand-alone PCC in compatibility mode for Windows 7 with the following operating system versions:

- Windows 8
- Windows Server 2012
- Windows 8.1
- Windows Server 2012 R2
- Windows 10
- Windows Server 2016

To activate the compatibility mode for the PCC, go to the *Properties* of the pcc.exe. In tab *Compatibility*, enable the option *Run this program in compatibility mode for* and select *Windows 7* in the drop-down menu.

Configuration of the stand-alone PCC (pcc.ini)

Communication from the PCC to the HYDRA server: Make the following settings in the local configuration file *pcc.ini* in the PCC directory (the rows with leading semicolon are comments):

[GateWay-Communication]

```
; Deactivate the connection "AIP8.2 -> PCC"  
Active=false
```

[HOST]

```
; Deactivate the connection "PCC -> AIP8.2"  
Active=0
```

[WSK]

```
; Configuration of the PDM-connection to the HYDRA server  
Host=<Host name or IP address of the HYDRA server >  
User=<Terminal number of the PCC according to MOC configuration>
```

[Server-Communication]

```
; Configuration of the communication to the HYDRA server via EVCOM  
Active=1
```

Important: Assign the terminal number in the hyusr.dat in the HYDRA server to the correct system.



For further information on the configuration of the PCC and the possibility to install the PCC as service, refer to the document [MBL_HYD-PCC.pdf](#).

Configuration of the AIP 8.2

If you use an AIP 8.2 for the data display and the manual input, make the following configurations or check these configurations:

MOC: Terminal configuration

In the [Terminal configuration](#), deactivate the option "Operated as MDE terminal" for the terminal.

MOC: Assignment of workplaces/machines to terminals

For all assigned workplaces, you must **not** set the option *Processing to MDE processing*.

Configuration ctaip.ini

Deactivate the direct AIP communication in the *ctaip.ini* of the AIP 8.2. To do so, comment out the following configuration (precede the rows by a semicolon):

```
; [DLL]  
; BusDLL=PCC.EXE
```

Activate the gateway communication:

[GateWay-Communication]

```
Active=true
```

Configuration pcc.ini

Activate the MDE blade in section Blades. To do so, make the following configuration (delete semicolons preceding the rows):

[BLADES]

BLADE_1=. \blades\mdeb2.dll

Restrictions when using the centralized MDE

Matrix for the use of the centralized collection

Which product group is available and can be used with a centralized collection and what are the restrictions:

Product group	Available with centralized collection	Restrictions
BDE	Yes	Up to CTAIP version 8.2.1.35, the function "Production lock" is not available on terminals that are only configured with "Operated as BDE terminal".
MDE	Yes	The function "Entry of disturbance reason required" is not supported.
PDV/EMG	Yes	
DNC	No	DNC is not supported with a centralized collection.
Scripting	Yes	From JAN 2019, also the access to the control (machine control) is available with the relevant functions in the terminal scripts. (for example, GetVal, SetVal for setting outputs, customer-specific connection of weighing machines and transfer of setting data to a machine when operation is logged on).

Product groups, which do not have an interface to the PCC, are not included in the matrix.

General restrictions

- If different clients are used (e.g. SMA and AIP), overlapping messages are possible.

Troubleshooting

When starting AIP 8.2, also the PCC is started although installed stand-alone.

Symptom

When you restart the AIP 8.2, the PCC is automatically started (You can see the icon in the taskbar) although it is installed stand-alone.

Solution

Check in the ctaip.ini if there is a further entry BusDLL=PCC.EXE which is not commented out. Delete the complete section [DLL].

Check in the [Terminal configuration](#) if the option "Operated as MDE terminal" is actually disabled.

Restart the terminal.

How do you know whether the centralized MDE is activated for a machine?

The centralized MDE is active, if the following requirements are fulfilled:

- In the MOC terminal configuration, a (separate) terminal configuration is available for the PCC.
- The license AIP-EBM and/or SCS-PCB is available.
- The MOC INI configuration includes an entry for the machine:

Name	MDE
Section	MQTT_NOTIFICATION
Key	<workplace number according to configuration>
Value	TRUE
Active	☒
Comment (optional)	

How do I know during upload whether the PCC runs in stand-alone operation?

The stand-alone PCC writes the log file: log_pdm.txt.

How do I know that the stand-alone PCC has connected to the MQTT Broker?

The stand-alone PCC writes the log file: log.mqtt.pcc.txt. The log file does not include any errors.

How do I know that the AIP has connected to the MQTT Broker?

The AIP writes the log file: log.mqtt.aip.txt. The log file does not include any errors.